EXPERIMENT NO : 3

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Develop and compare CLI, GUI, and Voice User Interfaces (VUI) for the same task and assess user satisfaction.

1.COMMAND LINE INTERFACE:

PYTHON CODE :

```
tasks = []


def add_task():
    task = input("Enter task: ")
    tasks.append(task)
    print("Task added successfully!")


def view_tasks():
    if not tasks:
        print("No tasks available.")
    else:
        print("\nTask List:")
        for i, task in enumerate(tasks, start=1):
            print(f"{i}. {task}")


while True:
    print("\n1. Add Task")
    print("2. View Task List")
```

```
    print("3. Exit")

    choice = input("Enter choice: ")

    if choice == "1":
        add_task()
    elif choice == "2":
        view_tasks()
    elif choice == "3":
        break
    else:
        print("Invalid choice!")
```

OUTPUT :

```
PS C:\Users\SABARI> cd "C:\Users\SABARI\OneDrive\Documents\uiux assignments\ex 3"
PS C:\Users\SABARI\OneDrive\Documents\uiux assignments\ex 3> py "cli ex 3.py"

1. Add Task
2. View Task List
3. Exit
Enter choice: 1
Enter task: assignment
Task added successfully!

1. Add Task
2. View Task List
3. Exit
Enter choice: 2

Task List:
1. assignment

1. Add Task
2. View Task List
3. Exit
Enter choice: 3
PS C:\Users\SABARI\OneDrive\Documents\uiux assignments\ex 3>
```

2.GRAPHICAL USER INTERFACE:

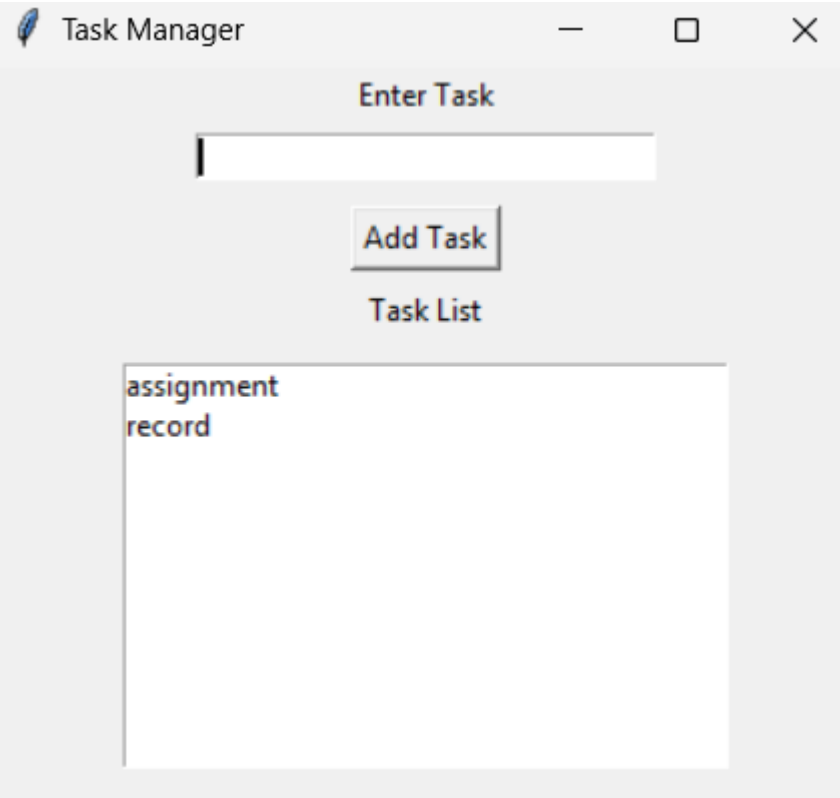
PYTHON CODE:

```
import tkinter as tk
from tkinter import messagebox

tasks = []

def add_task():
    task = task_entry.get().strip()
    if task == "":
        messagebox.showwarning("Warning", "Enter a task")
        return

    tasks.append(task)
    task_entry.delete(0, tk.END)
    update_list()

def update_list():
    task_listbox.delete(0, tk.END)
    for task in tasks:
        task_listbox.insert(tk.END, task)

root = tk.Tk()
root.title("Task Manager")
root.geometry("350x300")
```

```
tk.Label(root, text="Enter Task").pack()

task_entry = tk.Entry(root, width=30)
task_entry.pack(pady=5)

tk.Button(root, text="Add Task", command=add_task).pack(pady=5)

tk.Label(root, text="Task List").pack()

task_listbox = tk.Listbox(root, width=40)
task_listbox.pack(pady=10)

root.mainloop()
```

OUTPUT :

```
PS C:\Users\SABARI\OneDrive\Documents\uiux assignments\ex 3> py "gui ex 3.py"
```



3.VOICE USER INTERFACE:

PYTHON CODE:

```
import speech_recognition as sr

tasks = []

def process_command(command):
    command = command.lower()

    # ADD TASK
    if "add task" in command:
        task = command.replace("add task", "").strip()
```

```
        if task:
            tasks.append(task)
            print("✅ Task added:", task)
        else:
            print("⚠️ Please say task name")

    # VIEW TASKS
    elif "view tasks" in command:
        if not tasks:
            print("No tasks available")
        else:
            print("\nTask List:")
            for i, task in enumerate(tasks, 1):
                print(f"{i}. {task}")

    # EXIT
    elif "exit" in command:
        print("Exiting program...")
        return False

    else:
        print("Command not recognized")

    return True

def listen_voice():
```

```
    recognizer = sr.Recognizer()

    try:
        with sr.Microphone() as source:
            print("\n🎤 Speak command (Example: add task study / view
tasks)")
            recognizer.adjust_for_ambient_noise(source)
            audio = recognizer.listen(source)

        command = recognizer.recognize_google(audio)
        print("Voice Command:", command)
        return command

    except:
        print("⚠ Voice not detected. Switching to CLI input.")
        return input("Type command: ")

# MAIN LOOP
running = True
while running:
    command = listen_voice()
    running = process_command(command)
```

OUTPUT :

```
PS C:\Users\SABARI\OneDrive\Documents\uiux assignments\ex 3> py "vui ex 3.py"
⚠️ Voice not detected. Switching to CLI input.
Type command:
```

4.USER SATISFACTION COMPARISON:

PYTHON CODE:

```
def survey():
    print("Rate your satisfaction (1-5)")

    try:
        cli = int(input("CLI: "))
        gui = int(input("GUI: "))
        vui = int(input("VUI: "))

        if 1 <= cli <= 5 and 1 <= gui <= 5 and 1 <= vui <= 5:
            avg = (cli + gui + vui) / 3

            print("\nRatings:")
            print("CLI:", cli)
            print("GUI:", gui)
            print("VUI:", vui)
            print("Average Score:", avg)

        else:
            print("Enter values between 1 to 5")

    except:
```

```
        print("Invalid Input")


survey()
```

OUTPUT :

```
PS C:\Users\SABARI>  cd "C:\Users\SABARI\OneDrive\Documents\uiux assignments\ex 3"
PS C:\Users\SABARI\OneDrive\Documents\uiux assignments\ex 3>  py "user satisfaction.py"
Rate your satisfaction (1-5)
CLI: 4
GUI: 4
VUI: 4

Ratings:
CLI: 4
GUI: 4
VUI: 4
Average Score: 4.0
PS C:\Users\SABARI\OneDrive\Documents\uiux assignments\ex 3>
```